

INSTRUCTOR GUIDE

Product Management for Physical + Digital Systems

Course 03 — Facilitation Guide for Graduate Assistants and Faculty

Patrick T. Hoffman · @arcktip · patrickthoffman.com

HOW TO USE THIS GUIDE

This guide supports graduate teaching assistants and faculty facilitating Course 03: Product Management for Physical + Digital Systems. Each session entry provides a Session Overview, Theoretical Frame, Discussion Arc with timing, numbered Discussion Questions for all students (with facilitation notes), Graduate Questions (G1–G2, teal-accented), Common Misconceptions, Cross-session Connections, and Facilitator Notes.

The course is designed for a dual-listed seminar (upper-division undergraduate + master's) drawing students from urban planning, public policy, MBA, and engineering programs. The graduated question structure makes the two levels separable if needed.

Patrick's practitioner voice in this course

Course 03 is grounded in Patrick Hoffman's hands-on experience building physical-digital products — from Social Bicycles (SoBi) and the product that became JUMP, to his engagement with Superpedestrian's Copenhagen Wheel ecosystem, to product leadership at Google and Meta. Week 5 (the Copenhagen Wheel case) is the session most directly tied to his professional arc. His @arcktip frameworks on systems thinking, dependency mapping, firmware PM, and hardware roadmapping are the primary practitioner texts throughout the course. Where possible, Patrick joins Week 5 as a guest practitioner.

Final pitch evaluation criteria — share at Week 1

Students build toward these criteria from the first session. The final pitch must address all six:

- Hardware systems understanding — does the pitch show the seven ways hardware differs from software?
- BOM and margin model — is there a credible BOM estimate with a realistic margin waterfall?
- Dependency map and roadmap — is the product architecture visible with explicit decision gates?
- Build cycle strategy — does the team know what EVT/DVT/PVT will test and what constitutes a gate pass?
- Supply chain and firmware plan — CM selection, lead time constraints, and OTA strategy addressed?
- EOL plan — minimum service commitment, right-to-repair stance, and e-waste plan included?

Week 1 · Why Hardware Is Different

FORMAT

Lecture + case

TIMING

75 min

LEVEL

MBA + Grad Engineering

SESSION OVERVIEW

The opening session exists to unsettle students who arrive from software PM backgrounds — and to orient those coming from engineering who may not have product management vocabulary. The core argument is simple: every principle of software product management still applies to hardware, and none of them applies in the same way. Iteration is slower. Decisions are irreversible. Unit economics are inverted. Failure is physical. The session uses Nest as the primary case — not because it succeeded, but because it is the most fully documented physical-digital product that students can hold in their hands and trace from industrial design decision to margin outcome.

THEORETICAL FRAME

Cagan's Inspired is assigned selectively: students read it for the software PM baseline, then spend the session cataloging what hardware violates. The Juicero case (Week 3) is previewed but not yet assigned — use it as a teaser. The @arcktip Tetris epigraph ('not thinking ahead deeply is like playing Tetris') sets the systems thinking frame that Week 2 will develop formally.

SUGGESTED DISCUSSION ARC

0–10 min

Open with a physical object: hold up a smartphone, a Kindle, or if possible a Copenhagen Wheel hub motor. 'I want you to spend 60 seconds thinking about everything that had to go right before this object could exist in my hand. Not what it does — what it required.' Collect responses: components, manufacturing, firmware, certification, packaging, logistics, support.

10–25 min

Brief lecture: the seven ways hardware PM differs from software PM. (1) Iteration cycles are measured in months, not sprints. (2) Decisions are often irreversible — a wrong tooling choice costs \$80,000. (3) Unit economics are inverted — software is near-zero marginal cost; hardware cost is real per unit. (4) The product exists before the software is done. (5) Field reliability is life-or-death for margin. (6) Supply chain is a product dependency. (7) End-of-life has physical consequences.

25–50 min

Nest case discussion. Questions 1–3. The Nest case is used because it has unusually complete documentation: hardware, software, and business model decisions are all visible.

50–65 min

Graduate section: Ulrich/Eppinger, Anderson, Blank. Questions G1 and G2. The customer development question (G1) is where engineering students most frequently have to recalibrate.

65–75 min

Close: each student writes the product they will use for their PRD and final pitch. Keep these — use them Week 2 when students build their first dependency map.

DISCUSSION QUESTIONS — ALL STUDENTS

- 1** Nest's Learning Thermostat succeeded commercially. But it required \$150M in funding before shipping a single unit, and the first version had significant firmware bugs that required OTA fixes. What does that development arc tell you about the risk profile of hardware products compared to software products?

Push students to quantify: a bug in a mobile app costs the price of a developer sprint to fix. A bug in a shipped hardware product that requires a physical recall costs — what? Find the math.

- 2** The Nest thermostat's industrial design was a significant competitive advantage: it was the first 'beautiful' thermostat. How does industrial design function differently as a product constraint when you're building hardware vs. software?

Industrial design in hardware locks in manufacturing tooling — a beautiful product requires expensive custom tooling that becomes a sunk cost. In software, redesign is a sprint. In hardware, it's a new NRE charge.

- 3** Cagan argues that the best tech products are built by empowered, dedicated, durable product teams. Does that principle apply to hardware products? What changes about team composition, structure, and rhythm when the product has physical components?

Hardware teams necessarily include mechanical engineers, industrial designers, firmware engineers, supply chain managers, and certification specialists — roles that rarely appear in a pure software product team. The coordination overhead is real.

GRADUATE / MASTER'S QUESTIONS

- G1** Blank's customer development process — designed for software startups — assumes that you can build and test cheaply, learn quickly, and pivot freely. What modifications would you make to that process for a hardware startup? What can you prototype cheaply in hardware, and what cannot be prototyped?

- G2** Ulrich and Eppinger's product design process describes a phase gate structure: concept development, system-level design, detail design, testing and refinement, production ramp-up. How does that structure relate to Cagan's continuous discovery model? Are they compatible?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students from software backgrounds often assume that hardware iteration is slower because hardware engineering is less skilled or less agile. The actual constraint is physics and cost: tooling is expensive, components have long lead times, and physical testing takes time. Agile ceremonies don't change the speed of light.

MBA students sometimes treat hardware as a cost center and software as where all the value lives. Nest's acquisition for \$3.2B was primarily for the hardware product and the physical install base, not for the software. Hardware creates durable physical relationships with customers.

Graduate engineering students may know the technical constraints well but underestimate the PM role. The session should establish: the PM is not the engineer, is not the designer, and is not the marketer — but must be fluent in all three to make good decisions.

CONNECTIONS TO OTHER SESSIONS

Every subsequent session will return to the seven ways hardware differs from software introduced here. Post each on the board and keep them visible. In Week 3 (hardware economics), revisit #3. In Week 6 (firmware), revisit #4 and #5. In Week 7 (supply chain), revisit #6. In Week 14 (sunset), revisit #7.

FACILITATOR NOTES

Bring a physical object to every session of this course — the best hardware PM instructors teach with objects in hand. If you can source a Copenhagen Wheel hub motor (or any visible hub motor), Week 1 and Week 5 gain enormously. If not, a disassembled consumer electronics product (a broken Nest thermostat, an old iPhone with the back removed) works well. The goal is to make the atoms visible and real.

Week 2 · Systems Thinking

FORMAT

Lecture + lab

TIMING

90 min

LEVEL

MBA + Grad Engineering

SESSION OVERVIEW

Systems thinking is the course's foundational analytical tool. This session teaches it formally using Meadows's framework, then immediately applies it to hardware product architecture through the dependency mapping exercise that produces the Week 4 deliverable. The session is deliberately hands-on: students leave with a working draft dependency map for their own product concept, not just vocabulary.

THEORETICAL FRAME

Meadows's 'Thinking in Systems' is unusually accessible for a systems theory text — Donella Meadows was a science journalist before becoming a systems theorist, and her examples are drawn from everyday experience. The key concepts for hardware PM: stocks and flows (inventory is a stock; component consumption is a flow), feedback loops (positive feedback in viral adoption; negative feedback in quality control), and delays (the 18-month lead time for a hardware component is a delay that makes the system hard to control).

SUGGESTED DISCUSSION ARC

0–15 min

Brief lecture: Meadows's three core systems concepts. (1) Stocks and flows: what accumulates (inventory, technical debt, customer trust) and what changes it (shipments, bugs filed, support tickets closed). (2) Feedback loops: reinforcing (viral growth, price wars) and balancing (quality gate catches a defect before it ships). (3) Delays: the most important concept for hardware PM. A six-month component lead time means the system responds to decisions made six months ago — not decisions made today.

15–30 min

Apply to a familiar product: the iPhone. Draw the stocks and flows for an iPhone product launch on the whiteboard. What stocks are there? (Component inventory, finished goods, channel inventory, customer installs, support tickets.) What flows? (Production rate, ship rate, sell-through, returns.) What feedback loops? (Satisfaction → reviews → demand → production → inventory.) What delays? (Component lead times, firmware development, regulatory certification.)

30–60 min

Lab: students begin their own dependency map for their product concept. Instructor circulates. The map should show hardware layers (mechanical, PCB, battery), firmware/software layers (embedded firmware, mobile app, cloud backend), and dependency arrows with estimated lead times. Questions 1–3 weave through the lab.

60–80 min

Graduate section: Senge (archetypes), Bass/Clements (software architecture). Questions G1 and G2.

80–90 min

Close: each student identifies the single longest lead time in their dependency map. 'That

delay is your product's heartbeat — everything else has to work around it.' This point returns in Week 4 (roadmapping).

DISCUSSION QUESTIONS — ALL STUDENTS

- 1 **Meadows argues that most systems problems — traffic jams, inventory crises, arms races — share a common structure: stocks that change slowly, flows that react to current conditions, and delays that make feedback misleading. Map that structure onto a hardware product launch gone wrong. What was the stock? What was the delay? What decision was made on the basis of outdated information?**
Classic hardware example: a product team orders components based on optimistic demand forecast (outdated signal), the components arrive six months later when demand has shifted, and they now have excess inventory (a stock that cannot be quickly consumed).
- 2 **In your dependency map, which layer — hardware, firmware, or cloud — has the longest delay between a decision and its observable consequence? What does that tell you about where the hardest product management decisions live?**
Almost always: hardware mechanical decisions have the longest delays (tooling → production → testing → ship). Firmware is faster but still measured in weeks. Cloud can be deployed in hours. The PM's planning horizon must match the longest delay.
- 3 **Senge describes a 'fixes that fail' archetype: a solution relieves a symptom but creates a side effect that eventually brings the problem back, often worse. Find an example of 'fixes that fail' in hardware product development.**
Classic examples: reducing BOM cost by substituting a cheaper component → higher field failure rate → increased support costs that exceed the BOM savings. Or: accelerating a hardware schedule by skipping DVT testing → field reliability issues → expensive recall.

GRADUATE / MASTER'S QUESTIONS

- G1 **Software architecture texts distinguish between architectural decisions (hard to reverse, affect many components) and design decisions (easier to reverse, affect fewer components). Apply that distinction to a physical-digital product. Which hardware decisions are architectural? Which are design? What is the PM's role in distinguishing between them?**
- G2 **Senge's 'shifting the burden' archetype describes a situation where a symptomatic solution becomes so well-established that the fundamental problem-solving capacity atrophies. In what ways does over-reliance on firmware OTA updates exhibit this archetype? What fundamental hardware decisions is OTA patching allowing product teams to defer?**

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often treat the dependency map as an organizational chart — a picture of who reports to whom. A dependency map is a technical artifact: it shows what components depend on what, what information flows between layers, and where delays create synchronization risks.

MBA students sometimes resist the systems vocabulary as overly academic. The practical reframe: every hardware PM disaster (Juicero, Theranos, the Boeing 737 MAX software) can be traced to a misunderstood feedback loop or an ignored delay. The vocabulary is not decoration.

Graduate engineering students may have strong systems intuitions but weak PM vocabulary. Check that they can translate between: 'the tooling takes 12 weeks' (engineer) and 'hardware architecture decisions made this week have a 12-week observation delay, which means we must commit to them before we have full market validation' (PM).

CONNECTIONS TO OTHER SESSIONS

The dependency map started in this session is submitted as a deliverable at Week 4. Revisit it explicitly in Week 6 (firmware adds a layer), Week 7 (supply chain is a set of flows with delays), and Week 9 (IoT adds the connectivity and cloud layers). By Week 12, students should be able to draw the full map of their product without looking at their notes.

FACILITATOR NOTES

Lab sessions require circulating actively. The dependency map exercise reveals who understands systems thinking and who doesn't within the first 10 minutes — the students who draw boxes and then stop are missing the arrows (flows, dependencies, delays). Redirect them: 'What does this component need from that one? How long does it take to arrive?' The tool is the thinking, not the deliverable.

Week 3 · Hardware Economics

FORMAT

Lecture + model workshop

TIMING

75 min

LEVEL

MBA + Grad Engineering

SESSION OVERVIEW

This session teaches the financial language of hardware products: BOM (Bill of Materials), yield rate, NRE (non-recurring engineering), gross margin at target volume, and the volume pricing imperative. These concepts are not optional enrichment for hardware PMs — they are the primary vocabulary in which hardware product decisions are made and justified. A PM who cannot read and challenge a BOM is a liability, not an asset, to a hardware team.

THEORETICAL FRAME

The Juicero case is the course's best negative example: a product with a \$700 BOM built for a \$400 price, funded by \$120M of venture capital that obscured the fundamental unit economics problem until it was too late. Students who understand BOM math can spot a Juicero early. The session's lab involves building a simplified BOM for their own product concept — not to get the numbers right, but to learn to ask the right questions.

SUGGESTED DISCUSSION ARC

0–10 min

Open with the Juicero reveal: show the Guardian article (April 2017) that revealed Juicero's \$400 juice machine could be squeezed by hand. Ask: 'What does that reveal about the product? Not about the squeeze — about the BOM.' Surface: the value proposition was in the connectivity and subscription, not in the mechanical press. The BOM was over-engineered for the value it delivered.

10–25 min

Lecture: BOM fundamentals. (1) BOM = the complete list of components in a product with quantities and unit costs. (2) Yield rate = the percentage of units that pass quality testing at each production stage — each percentage point of yield loss is a direct margin hit. (3) NRE = the fixed costs of tooling, molds, and engineering that must be amortized across the production volume. (4) The volume pricing imperative: component costs fall as volume increases, but only if you commit to volume ahead of time.

25–55 min

Workshop: students build a simplified BOM for their product concept using the @arcktip template. Instructor circulates. Goal: populate at least 10 line items, estimate costs at 1,000 units and 100,000 units, and calculate a rough gross margin at both volumes. Questions 1–3 weave through the workshop.

55–70 min

Graduate section: Shapiro/Varian, Pisano, Teece. Questions G1 and G2.

70–75 min

Close: each student states their product's 'volume cliff' — the minimum volume at which their BOM math works. 'Below that number, you are paying to ship product. Above it, you have a business.'

DISCUSSION QUESTIONS — ALL STUDENTS

- 1 **The Juicero BOM was estimated at \$400 for a product sold at \$699 (later \$399). At what gross margin does a hardware business become viable? What factors determine how much margin hardware needs compared to software?**
Hardware businesses typically target 40–60% gross margin at scale. At launch, margins are often negative. The question is whether the volume trajectory will get to margin-positive before capital runs out. Juicero's answer was no.
- 2 **Yield rate is a tax on production that most software PMs have never encountered. If a product has 15 components and each has a 99% first-pass yield, what is the yield rate for a fully assembled unit?**
 $0.99^{15} \approx 0.86$ — approximately 14% of units fail first-pass testing and require rework or scrap. Walk students through the arithmetic. Then: what does that mean for a product with 50 components? 100? The math makes the case for simplifying the BOM.
- 3 **NRE costs are fixed and must be amortized across production volume. If a product has \$500,000 in tooling NRE and targets 10,000 units, how much does that add to the per-unit cost? At 100,000 units? What does this tell you about the relationship between hardware product strategy and volume targets?**
\$50/unit at 10k; \$5/unit at 100k. The implication: hardware product strategy is fundamentally a bet on volume. A product with a large NRE that can only reach 10,000 units may be commercially unviable even with a strong BOM.

GRADUATE / MASTER'S QUESTIONS

- G1 Teece's 'Profiting from Technological Innovation' distinguishes between innovators who capture value from their inventions and imitators who appropriate it. In hardware markets, what determines whether the innovator or the contract manufacturer captures the value? How does Juicero's experience illustrate this?
- G2 Shapiro and Varian argue that switching costs create lock-in and that lock-in is the primary source of durable competitive advantage in technology markets. What switching costs does a hardware product create that a software product cannot? How does that change the economics of a hardware business over time?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often confuse COGS (cost of goods sold) with BOM. BOM is the component cost; COGS includes BOM plus manufacturing labor, overhead, and quality testing. The distinction matters because COGS is what determines gross margin, not BOM alone.

The 'target cost' approach — setting the BOM budget from the desired gross margin and working backward — is unfamiliar to most students. Most students think of cost as given and price as the variable. In hardware, both must be managed simultaneously.

Graduate engineering students sometimes resist the financial framing as 'not their job.' Establish clearly: the PM owns the business case for the product, which requires understanding the BOM. An engineer who doesn't understand BOM math cannot evaluate trade-offs between two component options.

CONNECTIONS TO OTHER SESSIONS

The BOM workshop produces inputs for the Week 11 hardware economics model deliverable. Save student work from this session. The Juicero case recurs in Week 14 (sunset — what do you do with \$120M of inventory when the product is discontinued?). The volume pricing imperative connects to Week 13 (platform strategy — Peloton's razor-and-blades model is a hardware economics play).

FACILITATOR NOTES

The BOM workshop requires a spreadsheet template. Distribute the @arcktip course template before class. Students who have never seen a BOM will need line-item examples: for a connected consumer device, the major categories are mechanical (enclosure, frame, fasteners), electronic (PCB, chip, battery, antenna), display and interface, packaging, and firmware/certification. Have a sample BOM from a public hardware teardown (iFixit does these for every major consumer product) ready to show.

Week 4 · The Product Roadmap

FORMAT

Lecture + deliverable review

TIMING

75 min

LEVEL

MBA + Grad Engineering

SESSION OVERVIEW

Hardware roadmapping is one of the hardest and most consequential skills in this course. The key challenge: you must commit to hardware decisions 12–18 months before you can observe their market consequences. That means the roadmap must carry explicit uncertainty, staged commitment, and pre-defined decision gates. The 'now/next/later' framework familiar from software PM still works, but the timescales are different and the cost of changing 'later' to 'now' is an order of magnitude higher.

THEORETICAL FRAME

The session reviews the Week 4 deliverable (dependency map) as a group — anonymized if needed — and uses it to ground the roadmapping discussion. Students have already identified their longest lead time (from Week 2). The roadmap must be built around that constraint, not around the feature wish list.

SUGGESTED DISCUSSION ARC

0–15 min

Dependency map review: show 2–3 anonymized maps from Week 2. As a group, identify: (1) the longest lead time in each map, (2) the decision that has the most downstream dependencies, (3) one thing the map reveals that the student hadn't articulated in words.

15–35 min

Lecture: hardware roadmap structure. (1) Now lane: what is committed to current BOM, current firmware, current manufacturing partner. Changes here are expensive. (2) Next lane: what is being designed and validated — EVT/DVT in progress. Changes here are possible but costly. (3) Later lane: what is in concept — changes are cheap here. (4) Decision gates: explicit criteria that must be met before moving from Later to Next to Now. Gate criteria include: customer validation, supply chain confirmation, regulatory clearance, and margin model validation.

35–60 min

Workshop: students draft a 3-lane hardware roadmap for their product concept. Questions 1–3 weave through the workshop. Instructor focuses attention on the decision gates — the most commonly skipped element.

60–75 min

Graduate section: Reinertsen, Ward et al., Wheelwright/Clark. Questions G1 and G2. The set-based design concept (Ward) is the most important and least-known idea in hardware PM.

DISCUSSION QUESTIONS — ALL STUDENTS

1

A hardware roadmap must be built around the longest lead time in the product's dependency map. For your product, what is that lead time, and what product decisions

must you commit to before you have full market validation? How do you make good decisions under that uncertainty?

The honest answer: you cannot fully validate before committing to hardware decisions. The PM's job is to make those decisions with the best available information and to design the product architecture to minimize the cost of being wrong.

2 Cagan argues that product roadmaps are fundamentally flawed because they conflate output (features to be built) with outcomes (customer problems to be solved). How does that critique apply to a hardware roadmap? Can you build a hardware roadmap that specifies outcomes instead of features?

Yes, but it requires specifying the decision gates in outcome terms: 'Before committing to current BOM configuration, we must validate that users in the target customer segment are willing to pay \$X and that the use-case we're designing for represents >30% of actual usage.'

3 What is the difference between a decision gate and a milestone in a hardware roadmap? Why does that distinction matter for a hardware PM?

A milestone is a date by which something is done. A decision gate is a criterion by which you decide whether to proceed. Hardware PMs who only think in milestones commit to schedules without understanding the conditions under which those schedules are valid.

GRADUATE / MASTER'S QUESTIONS

G1 Reinertsen argues that long queues are the enemy of product development flow — work waiting to be done is waste that creates delays and obscures where the real bottleneck is. Map the queue structure in a hardware product development process. Where do queues form? Which queue, if eliminated, would have the greatest impact on overall cycle time?

G2 Ward et al.'s 'Second Toyota Paradox' argues for 'set-based concurrent engineering': explore multiple design options simultaneously and delay decisions until the last responsible moment, rather than committing to a single design early and iterating. How would you apply this principle to the BOM decisions in your product concept?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often build hardware roadmaps that look like Gantt charts: a list of features with dates. A hardware roadmap is a commitment structure, not a schedule. The dates are less important than the decision gates and the dependency order.

The 'last responsible moment' concept (from Toyota) is frequently misunderstood as 'procrastinate on decisions.' It means: don't make decisions that commit expensive resources before you have the information those decisions require. Make them as late as possible without creating downstream delays.

Graduate students sometimes want to skip the roadmap exercise because 'we don't know enough yet.'

That's exactly the point — the roadmap should make explicit what you know, what you don't know, and what you need to learn before each decision gate.

CONNECTIONS TO OTHER SESSIONS

The roadmap deliverable (dependency map + 3-lane roadmap) is submitted this week. Students will return to and revise it for the Week 15 final pitch. In Week 8 (PRD review), the roadmap is part of the PRD. In Week 12 (enterprise hardware), the roadmap becomes a sales tool — enterprise customers need to understand where the product is going before committing to a procurement.

FACILITATOR NOTES

The dependency map review at the top of class requires advance preparation: review all submitted maps before class and select 2–3 that illustrate different strengths and weaknesses. Anonymize clearly. The most instructive maps are not the best ones — they're the ones that reveal a common mistake most clearly.

Week 5 · Case Study: The Copenhagen Wheel

FORMAT

Case discussion + guest

TIMING

90 min

LEVEL

MBA + Grad Engineering

SESSION OVERVIEW

The Copenhagen Wheel is the course's central case: a physical-digital product with deep roots in academic research (MIT Senseable City Lab), a tortuous path from prototype to commercial product (seven years from concept to shipping), a complex pivot from B2C to B2B, and a direct connection to Patrick Hoffman's professional arc. The case illustrates nearly every concept from Weeks 1–4 in practice: hardware economics, dependency mapping, firmware complexity, supply chain decisions, and the fundamental challenge of taking a research product to market.

THEORETICAL FRAME

The session's unusual structure — part case analysis, part practitioner conversation — requires more preparation than a standard case discussion. Students should come having read the @arcktip essay and the MIT technical paper, and should have prepared at least one question for Patrick (or the invited practitioner) about a specific moment in the Copenhagen Wheel's development where a decision was made under uncertainty.

SUGGESTED DISCUSSION ARC

0–10 min

If Patrick is joining: brief introduction and context. If not: 5-minute recap of the Copenhagen Wheel arc for students who haven't encountered it: 2009 Copenhagen concept → 2012 Superpedestrian founded → years of EVT/DVT/PVT → 2017 consumer launch → 2020 discontinuation → pivot to LINK scooter platform.

10–45 min

Case analysis using the course's frameworks. Apply in sequence: (1) Dependency map — what were the key layers and dependencies in the Copenhagen Wheel product? (2) BOM question — what hardware economics challenge does an all-in-one hub motor create? (3) Roadmap — what decision gates did Superpedestrian face, and which did they navigate well or poorly? Questions 1–3.

45–70 min

If Patrick is guest: open conversation — let students ask questions they prepared. If not: graduate discussion. Questions G1 and G2.

70–80 min

Design exercise: 'If you were the PM at Superpedestrian in 2018 — after the consumer Copenhagen Wheel launched but before the pivot to scooters was finalized — what would your next roadmap look like? What decision gates would you set? What would you need to learn before committing?'

80–90 min

Close: 'The Copenhagen Wheel is a product that required MIT-level engineering, took seven years to ship, and was discontinued three years after launch. Was it a failure? What does the product's arc teach us about what success means for a physical-digital product?'

DISCUSSION QUESTIONS — ALL STUDENTS

- 1** The Copenhagen Wheel's core innovation — a fully integrated hub motor with battery, sensors, and Bluetooth in a single sealed wheel — was also its primary manufacturing challenge. How does the product's industrial design decision (all-in-one) shape every downstream decision: BOM, yield rate, field repairability, and end-of-life?

The sealed design was elegant and differentiated, but made repair impossible (field failures required full unit replacement) and yield rate harder to achieve (a single component failure in a sealed unit condemned the whole assembly). The design decision had BOM and support cost consequences the team may not have fully modeled.

- 2** Superpedestrian received 10,000+ Kickstarter-style pre-orders and then took several years to deliver. What product management failures does that gap reveal? What should the pre-order campaign have required the product team to demonstrate before launching?

Pre-orders create a cash flow that obscures the readiness of the product. A hardware product that is ready to take pre-orders should have passed DVT and have a confirmed manufacturing partner. Taking pre-orders at the EVT stage is a common and costly mistake.

- 3** Superpedestrian pivoted from B2C (consumer Copenhagen Wheel) to B2B (LINK scooter fleet platform). What does that pivot reveal about the difference between what a hardware product does for a consumer and what it requires from a business operator? Which market was better suited to the product's actual capabilities?

B2B micromobility operators can absorb a higher unit cost, have professional maintenance capacity, and need the fleet management features that the Copenhagen Wheel's connectivity enabled. Consumer buyers wanted simplicity, aesthetics, and low price — a harder match for a technically complex product.

GRADUATE / MASTER'S QUESTIONS

- G1** Abernathy and Utterback describe a 'dominant design' transition: when one design configuration becomes the standard around which an industry coalesces, the basis of competition shifts from product innovation to process innovation. Was the Copenhagen Wheel competing for dominant design in the e-bike market? What would winning that competition have required?

- G2** Christensen argues that sustaining innovations — which improve existing products along the performance dimensions that mainstream customers value — are ultimately vulnerable to disruptive innovations that start in lower-margin, overlooked segments. Where does the Copenhagen Wheel sit in that framework? Who was the potential disruptor?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students sometimes read the Copenhagen Wheel's discontinuation as a simple failure. Superpedestrian's pivot to the LINK scooter platform, funded in part by the technology and team built

around the Wheel, represents a different kind of success: a research project that became a company, which developed manufacturing capability, which found a better market. The arc is not a failure narrative.

The 'seven years from concept to shipping' timeline is sometimes treated as evidence of poor execution. In hardware, seven years for a product with custom hub motor design, embedded firmware, regulatory certification, and manufacturing setup is not unusual. The comparison to software timelines is the misconception.

Graduate engineering students may over-focus on the technical innovations and under-analyze the business model. The Wheel's most important lessons are business model lessons, not engineering ones.

CONNECTIONS TO OTHER SESSIONS

The Copenhagen Wheel case connects directly to Patrick Hoffman's professional arc — the micromobility space he worked in with Social Bicycles/JUMP overlaps directly with Superpedestrian's market. The Week 13 platform strategy session uses Superpedestrian's LINK fleet as a case of hardware-as-platform. The Week 5 case also sets up the Week 7 supply chain discussion: the Wheel's China manufacturing relationship is an example of the CM partner decisions discussed there.

FACILITATOR NOTES

If Patrick joins: prepare him with three specific questions about the product arc, not the technology. The most valuable practitioner insight is about decisions made under uncertainty: when did the team commit to the consumer market? What would have changed the pivot timeline? How did the regulatory environment shape the product roadmap? If no practitioner guest: the @arcktip essay carries this session. Read it closely and prepare to facilitate a close reading of specific passages.

Week 6 · Firmware & Software Integration

FORMAT

Technical lecture + lab

TIMING

75 min

LEVEL

MBA + Grad Engineering

SESSION OVERVIEW

Firmware is the most misunderstood layer of a physical-digital product by PM and MBA students, and the most underestimated source of product risk. This session demystifies firmware — what it is, how it differs from cloud software, what the PM's responsibilities are with respect to it — and teaches the specific practices (OTA strategy, backwards compatibility, rollback planning) that distinguish mature hardware PM practice from naive application of software PM methods.

THEORETICAL FRAME

The key insight: firmware is software, but it runs on hardware that is already in the field and cannot be easily recalled. An OTA update that bricks a device at a customer site is not a bug to be patched in the next sprint — it is a product recall event with real-world consequences (devices that cannot operate, support costs, potential safety implications). The PM must understand this before approving any firmware release.

SUGGESTED DISCUSSION ARC

0–10 min

Open with a real event: the 2016 Samsung Galaxy Note 7 battery fires. Walk through what a PM would need to have known to prevent that outcome. Not the chemistry — the process: what QA gates should have caught the battery defect, what firmware telemetry might have provided early warning, what the recall and EOL cost actually was (~\$5.3B before tax write-off).

10–25 min

Lecture: firmware PM fundamentals. (1) Firmware vs. application software: firmware is compiled for a specific hardware target, runs on microcontrollers without a full OS, and must be validated on the actual hardware. (2) OTA (over-the-air) updates: how they work, what can go wrong (download failure, power loss mid-update, incompatible hardware revision). (3) Backwards compatibility: the firmware on devices in the field must run on hardware from production runs that may be 3 years old — component changes between runs create compatibility risk.

25–55 min

Lab: firmware dependency mapping. Students add the firmware layer to their existing dependency map from Week 2. For each firmware module, they must identify: (1) which hardware components it interfaces with, (2) which cloud services it depends on, (3) what happens if an OTA update is interrupted mid-process.

55–70 min

Graduate section: Ganssle, Raymond, Cusumano/Selby. Questions G1 and G2.

70–75 min

Close: 'OTA updates feel like a safety net — if you ship a bug, you can fix it later. What's the condition under which OTA fails as a safety net?' (Answers: customer has disabled auto-update, device is offline, OTA itself has a bug, hardware revision incompatibility.)

DISCUSSION QUESTIONS — ALL STUDENTS

- 1** An OTA update is pushed to 500,000 devices in the field. The update has a bug that bricks devices running on a specific hardware revision from 18 months ago. Approximately 30,000 devices are affected. What should the PM have required before approving the update for production rollout?

Required: regression testing on all hardware revisions currently in the field, a staged rollout protocol (push to 1% of devices first, monitor for 24 hours, then proceed), a rollback mechanism, and explicit acceptance criteria including a 'brick rate' threshold.

- 2** Backwards compatibility in firmware requires supporting the API contract between firmware and cloud backend for every hardware revision in the field. How does a hardware product PM manage that compatibility requirement? What happens when a cloud service must change in a way that breaks older firmware?

Best practice: maintain a versioned firmware API contract; deprecate old versions with advance notice to customers; create a firmware update campaign to move all devices to a current version before retiring old API support. The PM must track the version distribution of their fleet.

- 3** The firmware rollout strategy for a consumer product differs from that for a fleet product (B2B). For a consumer Copenhagen Wheel, what rollout strategy would you recommend? For a fleet of 10,000 shared scooters managed by a city operator?

Consumer: staged rollout by user cohort, opt-in for beta users, monitor crash telemetry. Fleet: coordinate with operator, schedule updates during low-usage hours, maintain a known-good firmware version for rollback, require explicit operator sign-off for critical firmware changes.

GRADUATE / MASTER'S QUESTIONS

- G1** Ganssle argues that most firmware quality problems are not caused by lack of skill but by schedule pressure that forces teams to skip testing steps. What organizational and product management practices protect firmware quality against schedule pressure? What is the PM's role in creating those protections?

- G2** Raymond's 'Cathedral and the Bazaar' describes two development models: hierarchical (cathedral) and open/distributed (bazaar). Which model does firmware development resemble, and why? What would it take to apply bazaar-style development to firmware for a commercial hardware product?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

The phrase 'we can fix it with a firmware update' is the most dangerous sentence in hardware PM. It is sometimes true, but it depends on: the device being online, the user having consented to updates, the update working on all hardware revisions in the field, and the fix not introducing new bugs. Treat OTA as a powerful but unreliable tool.

MBA students often don't distinguish between firmware and the mobile app that accompanies a hardware product. The mobile app runs on the customer's phone (relatively easy to update); firmware runs on the device (much harder to update safely). They have different PM rhythms, different risk profiles, and different testing requirements.

Graduate engineering students may underestimate the PM's role in firmware decisions. Firmware is where hardware economics, reliability, and customer experience all intersect. The PM who cannot articulate the rollout strategy for a firmware update is flying blind.

CONNECTIONS TO OTHER SESSIONS

Firmware connects to Week 9 (IoT — the connectivity layer that makes OTA possible), Week 11 (field reliability — firmware bugs are one of the primary causes of field returns), and Week 14 (sunset — end-of-life for a hardware product must include a plan for what happens to firmware when cloud services are discontinued). The Samsung example also previews the Week 14 discussion of obsolescence.

FACILITATOR NOTES

The Samsung Note 7 example is well-documented and works well as an opener because it is visceral and familiar. Other options: the Volkswagen emissions scandal (firmware as fraud), the Boeing 737 MAX MCAS software (safety-critical firmware in hardware context), or the Peloton treadmill recalls (sensor firmware and safety). Choose based on what is most current and most relevant to your student audience.

Week 7 · Supply Chain as Product Strategy

FORMAT

Lecture + negotiation simulation

TIMING

75 min

LEVEL

MBA + Grad Engineering

SESSION OVERVIEW

Supply chain decisions are product decisions: choosing a component supplier, selecting a contract manufacturer, deciding to near-shore or offshore production — these are choices that determine the product's cost structure, resilience, feature set, and ultimately its competitive position for years. Most hardware PMs learn this the hard way, when a sole-source component becomes unavailable (the 2021 semiconductor shortage affected nearly every hardware product company) or when a CM relationship sours and the alternative takes 12 months to qualify.

THEORETICAL FRAME

Fisher's 'functional vs. innovative product' framework from HBR is the most useful single lens for supply chain strategy: functional products (stable demand, long product lifecycles) need efficient supply chains (lean, low-cost, high-utilization); innovative products (unpredictable demand, short lifecycles) need responsive supply chains (flexible, excess capacity, faster replenishment). Hardware products, especially in consumer tech, are innovative products in Fisher's taxonomy — but are often managed with functional supply chain logic, producing mismatches that become crises.

SUGGESTED DISCUSSION ARC

0–10 min

Open with the 2021 semiconductor shortage as a live case. A chip that cost \$0.50 in 2020 cost \$5.00 in 2021 because of supply disruptions. Companies with sole-source dependencies on that chip had to pay 10x or halt production. Companies that had diversified supply were able to continue. 'What product decisions made in 2018 determined whether a hardware company survived 2021?'

10–25 min

Lecture: supply chain strategy for hardware PMs. (1) Sole-source vs. multi-source: the risk/cost trade-off. (2) Qualified supplier alternatives: the discipline of qualifying a backup supplier before you need them. (3) Component lead times as a product planning constraint: if your key processor has a 26-week lead time, your product planning horizon is at least 26 weeks. (4) Near-shoring and the resilience trade-off: manufacturing closer to your market reduces logistics risk but increases unit cost.

25–55 min

Negotiation simulation: pairs of students play a hardware startup PM (wants flexible minimum order quantities and short lead times) and a contract manufacturer sales rep (wants large, firm orders and long lead times). 10 minutes of negotiation, then debrief. Questions 1–3 weave through the debrief.

55–70 min

Graduate section: Fisher, Chopra/Meindl, Lee. Questions G1 and G2.

70–75 min

Close: students update their dependency maps with supply chain information — for each hardware component, add the supplier name, lead time, and whether an alternative

source has been qualified. 'Your supply chain is now visible in your dependency map.'

DISCUSSION QUESTIONS — ALL STUDENTS

- 1** The 2021 semiconductor shortage forced many hardware companies to choose between halting production and paying 5–10x spot market prices for components. What supply chain practices, adopted in 2019 or 2020, would have reduced that exposure? What is the PM's role in ensuring those practices are in place?

Practices: multi-source qualification for critical components, buffer inventory policy for long-lead items, contractual supply commitments with volume penalties, and product design choices that enable component substitution (designing for multiple compatible chips rather than a single sole-source one).

- 2** A contract manufacturer offers you a 30% reduction in unit cost if you commit to a minimum order of 50,000 units per quarter. Your current demand forecast is 20,000 units per quarter. What additional information do you need before accepting that offer? What are the risks of accepting it? Of declining it?

Need: confidence interval on demand forecast, capacity to finance and store 30,000 excess units per quarter, penalty terms for under-ordering, and assessment of what happens to margin if demand stays at 20,000. Risk of accepting: inventory accumulation. Risk of declining: higher unit cost that may make the product uncompetitive.

- 3** Near-shoring (moving manufacturing closer to the target market) reduces logistics risk and lead times but typically increases unit cost by 15–30%. Under what conditions does the risk reduction justify the cost increase? What product characteristics make near-shoring more or less valuable?

Near-shoring is more valuable for: products with short product lifecycles (where demand unpredictability is high), products with high warranty costs (where field failures require fast parts supply), and products where regulatory requirements mandate domestic manufacturing. Less valuable for: commodity hardware with stable demand and low service intensity.

GRADUATE / MASTER'S QUESTIONS

- G1** Fisher argues that the right supply chain for an innovative product is fundamentally different from the right supply chain for a functional product — and that the most common supply chain mistake is applying functional logic to innovative products. Classify your product concept using Fisher's taxonomy. Does your current supply chain approach match that classification?

- G2** Lee's 'Triple-A Supply Chain' (agile, adaptable, aligned) describes practices that make supply chains resilient to disruption. Which of Lee's three As is hardest to achieve for a hardware startup? Which is most important during the first year of production ramp?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often think of supply chain as an operations function that 'the operations team handles.' In hardware, supply chain decisions are product decisions — they determine what features are possible, what the product's cost will be, and when the product can be available. The PM must own the supply chain strategy even if they don't manage the relationships day-to-day.

The phrase 'we'll figure out supply chain when we get to manufacturing' is one of the most expensive mistakes in hardware product development. By the time you're at manufacturing, you've locked in component choices that determine your supply chain options. Supply chain strategy begins at the BOM.

Graduate engineering students sometimes treat sole-source components as acceptable because the component is technically superior. The PM must challenge this: technical superiority has to be weighed against supply risk. A 10% better component that creates a sole-source dependency may not be worth the exposure.

CONNECTIONS TO OTHER SESSIONS

Supply chain connects to Week 3 (BOM — the BOM determines your supplier list), Week 4 (roadmap — lead times are the hardest constraints on the hardware roadmap), Week 10 (contract manufacturers — the supply chain session introduces CM relationships; Week 10 goes deep on the EVT/DVT/PVT process), and Week 14 (sunset — supply chain discontinuations are one of the primary triggers of product EOL).

FACILITATOR NOTES

The negotiation simulation requires role cards prepared in advance. The key asymmetry to build in: the startup PM has strong market demand signals but weak balance sheet; the CM rep has excess capacity but needs order commitment. The natural resolution — a smaller minimum order in exchange for a cost premium — is the 'right' answer, and most student pairs find it. The debrief should focus on what information each side revealed and what they withheld.

Week 8 · Midterm: PRD Review**FORMAT**

Presentations + structured critique

TIMING

90 min

LEVEL

MBA + Grad Engineering

SESSION OVERVIEW

The midterm PRD review is a skills assessment disguised as a critique session. Students present their draft Product Requirements Documents — not polished deliverables but working drafts — for structured critique on completeness, internal consistency, technical feasibility, and market grounding. The instructor's role is to model rigorous PRD critique: every claim in a PRD should be either a testable specification or an explicitly stated assumption.

THEORETICAL FRAME

A good hardware PRD differs from a software PRD in several important ways: it must include hardware-specific sections (BOM targets, regulatory requirements, physical form factor constraints, manufacturing tolerances) alongside the standard sections (user stories, acceptance criteria, non-functional requirements). A PRD that could describe any product has not done enough work.

SUGGESTED DISCUSSION ARC**0–10 min**

PRD critique framework: introduce the four questions that should be asked of every claim in a PRD. (1) Is this a specification (testable) or a wish (untestable)? (2) Is this hardware-constrained, firmware-constrained, cloud-constrained, or unconstrained? (3) What happens to this requirement if the BOM cost target is missed by 20%? (4) How does this requirement change if the product launches 6 months late?

10–75 min

Presentations: each student presents for 8 minutes, receives 12 minutes of structured critique. Critique covers: completeness (are all sections present?), specificity (are requirements testable?), hardware grounding (does the PRD acknowledge physical constraints?), and feasibility (does the roadmap support the requirements?).

75–90 min

Plenary synthesis: 'Across all PRDs, what was the most common gap? What was the strongest section? What section is most commonly written as if this were a software product?'

DISCUSSION QUESTIONS — ALL STUDENTS**1**

A PRD requirement reads: 'The product shall have long battery life.' What is wrong with this requirement, and how would you rewrite it as a testable specification?

'Long' is not testable. Rewrite: 'The product shall operate for a minimum of 8 hours of continuous use at 25°C ambient temperature from a single charge, tested at the 5th percentile of the field battery performance distribution across 1,000 production units.'

- 2** Your PRD specifies a target BOM of \$45. During the critique, an engineer points out that the wireless connectivity module alone costs \$22 at your target volume. What does that single data point require you to do to the rest of the PRD?

It requires reconsidering either (1) the BOM target (and therefore the margin model), (2) the connectivity specification (can you use a lower-cost protocol?), or (3) the target price (can you charge enough to support a \$45+ BOM?). The PRD must be internally consistent.

- 3** Your PRD does not mention regulatory certification requirements. An instructor asks: what certifications will this product require? What should your PRD say about certification, and when in the roadmap do certification requirements create constraints?

For a consumer electronics product in the US: FCC Part 15 (radio frequency emissions), UL or ETL (electrical safety), potentially FCC ID for Bluetooth/WiFi. Certification requirements affect the hardware design (antenna placement, EMI shielding) and the schedule (FCC testing can take 8–12 weeks). The PRD should identify required certifications and note their design implications.

GRADUATE / MASTER'S QUESTIONS

- G1** Ulrich and Eppinger's product specifications process moves from customer needs to target specifications, then to final specifications after feasibility testing. Does your PRD follow that structure? At what point in your development process will your draft specifications become final specifications, and what evidence will trigger that commitment?

- G2** A PRD is a contract between the PM, the engineering team, and the business. It sets expectations that will be evaluated at launch. What should a hardware PM do when market conditions change after the PRD is finalized but before the product ships? What is the change management process for a hardware PRD?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often confuse requirements with solutions. 'The product shall use Bluetooth 5.2' is a solution specification; 'the product shall wirelessly connect to a paired smartphone within 3 seconds under normal indoor conditions' is a requirement. Requirements describe what the product must do; solutions describe how it does it. The PM owns the requirements; the engineering team owns the solutions.

A long PRD is not a good PRD. The best PRDs are precise, not comprehensive — they specify exactly what matters and acknowledge explicitly what is not yet decided. Vague requirements that appear complete are worse than explicit gaps.

Graduate engineering students sometimes write PRDs that read like technical design documents. The PRD is not the engineering spec — it is the PM's statement of what the product must achieve and within what constraints. The engineering spec is derived from the PRD by the engineering team.

CONNECTIONS TO OTHER SESSIONS

The PRD is a living document that evolves through the second half of the course. After the midterm critique, students revise their PRDs based on feedback and incorporate new content from Weeks 9–14 (IoT requirements, manufacturing constraints, enterprise requirements if applicable, EOL plan). The final pitch (Week 15) includes the revised PRD as a deliverable.

FACILITATOR NOTES

The PRD critique should be genuinely rigorous — not cruel, but demanding specificity. The four questions in the arc (testable? constrained? cost-resilient? schedule-resilient?) work well as a consistent framework. Apply them to every PRD: 'This requirement — is it testable as written? What would the test look like?' Students who are challenged in this way leave with a much sharper sense of what a hardware PRD must accomplish.

Week 9 · IoT & Connectivity

FORMAT

Technical lecture + lab

TIMING

75 min

LEVEL

MBA + Grad Engineering

SESSION OVERVIEW

This session provides the PM-level understanding of IoT architecture that the course requires — not sufficient to implement a connectivity solution, but sufficient to make informed decisions about connectivity protocols, edge vs. cloud trade-offs, data collection strategy, and the security implications of always-on hardware. Students who leave this session should be able to participate meaningfully in a technical review of their product's connectivity architecture.

THEORETICAL FRAME

The landscape of IoT connectivity protocols (WiFi, Bluetooth Low Energy, Zigbee, Z-Wave, LTE-M, NB-IoT, LoRaWAN) is genuinely complex, and students should not try to memorize it. The PM's job is to understand the trade-off dimensions — power consumption, range, bandwidth, cost, latency — and to map their product's requirements to the appropriate protocol. The session's lab forces that mapping.

SUGGESTED DISCUSSION ARC

0–10 min

Open with a connectivity failure: the Nest Learning Thermostat's WiFi connectivity issues in homes with enterprise-grade network security (corporate networks, university housing). 'A PM who didn't understand that Nest's WiFi implementation would fail on 802.1X networks shipped a product that didn't work in a large market segment. What would understanding IoT connectivity have prevented?'

10–30 min

Lecture: IoT connectivity decision framework. (1) The trade-off triangle: power consumption vs. range vs. bandwidth. No protocol is best on all three. (2) Protocol overview: BLE (low power, short range, low bandwidth — wearables, beacons), WiFi (high power, medium range, high bandwidth — home devices), LTE-M/NB-IoT (cellular, medium power, wide range, low bandwidth — infrastructure, fleet), LoRaWAN (very low power, very long range, very low bandwidth — sensors, agriculture). (3) Edge vs. cloud: what computation happens on the device vs. in the cloud, and what are the latency, cost, and privacy implications.

30–55 min

Lab: students select a connectivity protocol for their product concept and justify the choice using the decision framework. They must also answer: (1) what data does the product collect, (2) where is it stored, (3) who can access it, and (4) what are the privacy implications? Questions 1–3 weave through the lab.

55–70 min

Graduate section: Zuboff, Swan, Rose. Questions G1 and G2.

70–75 min

Close: add the connectivity layer and data flows to the dependency map. 'Your product's IoT architecture is now visible.'

DISCUSSION QUESTIONS — ALL STUDENTS

- 1 **A fitness tracker uses Bluetooth Low Energy to sync data to a companion app. A fleet management platform uses LTE-M to track vehicle locations continuously. These are both IoT products with very different connectivity requirements. Build the decision framework: what questions should a hardware PM ask to determine which protocol is right for their product?**
Questions: (1) Does the product need to be always-connected or sync periodically? (2) What is the power budget — battery-powered or plugged in? (3) What is the data volume — kilobytes per day or megabytes? (4) What is the acceptable latency — milliseconds or hours? (5) Will the product be in WiFi-covered environments or in the open field?
- 2 **Schneier argues that the security of IoT devices is consistently sacrificed for cost and time-to-market, creating systemic vulnerabilities in critical infrastructure. From a PM perspective, what security requirements should be non-negotiable for any connected hardware product? Who should enforce those requirements, and at what point in the development process?**
Non-negotiables: encrypted communication (TLS), authenticated device identity (device certificates), secure boot (firmware integrity verification), and a defined patch lifecycle (how long will security updates be provided?). These must be specified in the PRD and tested at DVT.
- 3 **A product collects location data, usage patterns, and health metrics from 500,000 connected devices. The PM wants to use this data to improve the product and to license it to third parties. What product management decisions must be made before either use is permissible? What changes if users are in the EU (under GDPR)?**
Required before using data: explicit user consent for each use category, data minimization (collect only what is needed), data retention limits, and a privacy policy that accurately describes what is collected and why. GDPR adds: right to erasure, data portability, and mandatory breach notification within 72 hours.

GRADUATE / MASTER'S QUESTIONS

- G1 Zuboff's surveillance capitalism framework describes a business model in which user data is the product. How does that framework apply to a hardware product company? What is the difference between using fleet data to improve the product and using it as a surveillance capitalism revenue stream?
- G2 Rose's 'Enchanted Objects' describes a design philosophy of ambient intelligence — hardware that enhances the environment without demanding attention. How does that design philosophy constrain the connectivity and data collection architecture? What trade-offs does ambient intelligence make relative to app-centric design?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often assume WiFi is always the right choice because it's familiar and high-bandwidth. WiFi's power consumption makes it impractical for battery-powered devices, and its dependence on a home network infrastructure creates reliability issues in commercial and institutional deployments. The protocol choice must match the deployment environment.

'Security can be added later' is not true for hardware. Security features that require hardware support (secure element, hardware encryption accelerator) must be designed into the hardware before manufacturing. A security vulnerability that requires a hardware change to fix is not fixable in the field.

Graduate students sometimes treat IoT connectivity as a pure engineering decision. The PM must make four decisions about connectivity before engineering implements any of them: which protocol, what data to collect, where data is processed, and who has access. These are product decisions with business, legal, and ethical implications.

CONNECTIONS TO OTHER SESSIONS

IoT connectivity connects to Week 6 (firmware — OTA updates require the connectivity layer to work), Week 11 (field reliability — connectivity failures are a major source of field returns), Week 12 (enterprise hardware — enterprise IT departments have specific connectivity requirements), and Week 14 (sunset — cloud service dependencies must be managed at EOL).

FACILITATOR NOTES

The connectivity protocol overview is where engineering students will be most comfortable and MBA students most uncomfortable. Pair them deliberately in the lab: the engineering student knows which protocols exist; the MBA student forces the business justification for each choice. The combination produces better decisions than either student would make alone.

Week 10 · Manufacturing Partners

FORMAT

Process walkthrough + simulation

TIMING

75 min

LEVEL

MBA + Grad Engineering

SESSION OVERVIEW

Working with a contract manufacturer is one of the most consequential and least-taught skills in hardware PM. This session walks through the complete CM engagement lifecycle: from initial NDA and RFQ through EVT, DVT, and PVT build cycles, to mass production ramp. Students leave with a detailed understanding of what happens at each build cycle, what the PM is responsible for at each gate, and what the most common and costly mistakes are.

THEORETICAL FRAME

Chen's chapter on contract manufacturing is the most practical single reading in the course — it should be read closely, not skimmed. The session extends and applies it through simulation: students work through a fictional EVT build cycle, receiving a set of test results and deciding which findings constitute a gate pass, which require iteration, and which require a fundamental design change.

SUGGESTED DISCUSSION ARC

0–15 min

Lecture: the build cycle sequence. (1) POC (proof of concept): does the core technology work at all? Small quantities, often in-house. (2) EVT (engineering verification testing): does the design work in manufactured form? 50–100 units from the CM, test against engineering specs. (3) DVT (design verification testing): does the design work at near-production quality? 200–500 units, test against all product requirements. (4) PVT (production verification testing): does the production process work? Full production line, test yield rate and process consistency. (5) MP (mass production): ongoing production.

15–35 min

Walk through the EVT/DVT/PVT sequence for a specific product — the Copenhagen Wheel or a student's PRD concept. For each build cycle: what is being tested, who is responsible, what are the pass/fail criteria, and what happens if you fail?

35–60 min

Simulation: students receive an EVT build report (fictional) with 12 findings. Working in pairs, they must classify each finding as: (1) pass, (2) fix in DVT, (3) requires design iteration back to EVT, or (4) fundamental redesign required. Pairs share their classifications and reasoning. Questions 1–3 weave through the debrief.

60–75 min

Graduate section: Pisano/Shih, Utterback, Sturgeon. Questions G1 and G2. The Pisano/Shih 'industrial commons' argument is the most provocative — what happens to product innovation when manufacturing moves offshore?

DISCUSSION QUESTIONS — ALL STUDENTS

- 1** An EVT build report shows that 23% of units fail drop testing (the product is specified to survive a 1-meter drop). The engineering team says the issue is fixable in DVT by adding rubber bumpers to the corners. The industrial designer says rubber bumpers will compromise the product's aesthetic. How do you resolve this as a PM?

This is the canonical hardware PM dilemma: technical requirement vs. design intent. Resolution path: (1) quantify the cost of field failures due to drops vs. the revenue impact of the aesthetic compromise. (2) Run a conjoint analysis or user test to measure customer sensitivity to the aesthetic change. (3) Explore whether an internal structural change (hidden reinforcement) can meet the drop spec without exterior modification.

- 2** A DVT build is completed with 94% yield — 6% of units fail at final quality inspection. Your target yield for mass production is 98%. The CM says they can improve yield to 97% with process adjustments, but reaching 98% will require an additional \$200,000 in manufacturing engineering investment (NRE). How do you evaluate that investment?

At 200,000 units shipped, a 1% yield improvement saves approximately $2,000 \text{ units} \times (\text{unit BOM cost} + \text{rework cost})$ in waste. If that math exceeds \$200,000, the investment pays back in the first production run. The PM must do this calculation, not just accept the CM's assessment.

- 3** Your CM is in Shenzhen and has a 6-week lead time for your key assembly. A second CM in Vietnam offers equivalent quality and a 4-week lead time, but at 8% higher unit cost. Under what market conditions does 2 weeks of lead time reduction justify an 8% unit cost increase?

Two weeks of lead time reduction is worth: (1) reduced safety stock inventory, (2) faster response to demand changes, (3) reduced capital tied up in transit inventory. At high volume with volatile demand, 2 weeks can be worth more than 8% COGS. At stable demand and lower volume, probably not. The PM must model the specific case.

GRADUATE / MASTER'S QUESTIONS

- G1** Pisano and Shih argue that the offshoring of manufacturing has degraded the 'industrial commons' — the shared infrastructure of skills, suppliers, and process knowledge that enables product innovation. If that argument is correct, what does it imply for a hardware startup's decision about where to manufacture? Is there a version of this course that should recommend domestic manufacturing on strategic grounds?
- G2** Sturgeon describes contract manufacturing as a distinct industrial system — not just cheaper production but a fundamentally different organizational model for hardware production. What are the PM's obligations to the CM relationship? What does a healthy long-term PM/CM partnership look like, and how does it differ from a purely transactional relationship?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often think EVT failure means the product has failed. EVT is expected to fail — that is its purpose. The question is whether the failures are expected (known risks being measured) or unexpected (unknown risks appearing). Unexpected EVT failures are the ones that should alarm a PM.

The transition from DVT to PVT is often treated as a schedule milestone. It is actually a quality gate: the product is ready for PVT when the DVT build has met all acceptance criteria, not when the calendar says it should. PMs who push to PVT on schedule when DVT criteria haven't been met create expensive PVT failures.

Graduate engineering students sometimes think of the CM as a vendor to be managed at arm's length. The most successful CM relationships function as partnerships: the CM's process engineers have insights about manufacturing yield and component availability that significantly improve the product if the PM creates conditions for them to share it.

CONNECTIONS TO OTHER SESSIONS

Manufacturing connects to Week 7 (supply chain — the CM is part of the supply chain) and Week 11 (field reliability — yield rate at manufacturing determines the baseline field failure rate). The EVT/DVT/PVT sequence is referenced in the final pitch evaluation criteria: teams must specify at what build stage their product concept currently is.

FACILITATOR NOTES

The EVT build report simulation requires a prepared document. Create a fictional product (similar to student concepts but not identical to any of them) with 12 EVT findings that span all four classifications. The most instructive findings are the borderline ones: a finding that is 'fix in DVT' for one team might be 'fundamental redesign' for another, depending on their product's use case. The debrief should surface this ambiguity explicitly.

Week 11 · Field Reliability & Support

FORMAT

Lecture + economics model workshop

TIMING

75 min

LEVEL

MBA + Grad Engineering

SESSION OVERVIEW

This session is the most quantitative in the course and produces the hardware economics model deliverable. Field reliability is where hardware product decisions made in the first half of the development cycle have their most expensive consequences: a 2% field failure rate sounds small until you're shipping 500,000 units per year. Students leave with a working model that connects yield rate, field failure rate, return rate, support cost, and warranty cost into a coherent picture of hardware unit economics.

THEORETICAL FRAME

The session uses the @arcktip hardware economics model template as the primary working document. Students have already populated some inputs in Week 3 (BOM workshop). This session adds the reliability and support cost dimensions. The result is a model that shows the margin waterfall from BOM to gross margin after warranty — the picture that every hardware CFO and investor will ask for.

SUGGESTED DISCUSSION ARC

0–10 min

Open with the margin waterfall concept: show a simplified P&L for a hardware product that goes from \$149 retail price to negative gross margin after warranty and support costs. 'This is how a hardware product with a positive BOM margin becomes a business that is losing money on every unit shipped. Walk me through where the margin went.' Surface: field failure rate × replacement cost + support cost per ticket × tickets per unit = warranty and support expense.

10–25 min

Lecture: field reliability concepts. (1) MTBF (mean time between failures): the average time a unit operates before failing. A product with an MTBF of 5,000 hours that users run for 4 hours/day will see 1 in 10 units fail in the first year. (2) Field failure rate vs. yield rate: yield rate is failures at production; field failure rate is failures after deployment. Both affect total return cost. (3) The cost of a field return: not just the unit replacement cost but the return shipping, refurbishment or disposal, and the lifetime value impact of a customer who experienced a failure.

25–60 min

Workshop: students complete the hardware economics model for their product concept. Instructor circulates. Key outputs: (1) gross margin at target BOM, (2) warranty cost as a percentage of revenue, (3) break-even volume, (4) net margin after warranty at three volume scenarios.

60–75 min

Graduate section: Juran/Godfrey, Kano, Reinertsen. Questions G1 and G2.

DISCUSSION QUESTIONS — ALL STUDENTS

- 1** Your product has a 2% annual field failure rate and ships 200,000 units per year. Replacement units cost \$35 (your BOM cost). Return shipping costs \$8. Customer support costs \$12 per return contact. What is your annual warranty expense? What gross margin do you need to absorb that cost?

Annual failures: 4,000 units. Replacement: $4,000 \times \$35 = \$140,000$. Shipping: $4,000 \times \$8 = \$32,000$. Support: $4,000 \times \$12 = \$48,000$. Total: \$220,000. At 200,000 units $\times$ \$100 ASP = \$20M revenue, warranty expense is 1.1% of revenue. At lower ASP or higher failure rate, the math becomes more painful quickly.

- 2** The Kano model distinguishes between 'basic quality' (failure is noticed and harmful), 'performance quality' (more is better), and 'excitement quality' (unexpected, delightful). How does this model apply to hardware reliability? What is the basic quality threshold for a consumer electronics product?

Basic quality for consumer electronics: the product must work reliably under normal use conditions. Below this threshold, customers are not just dissatisfied — they are angry, and they write negative reviews that suppress demand. The specific threshold varies by product category and price point, but 1% annual failure rate is often cited as the threshold where word-of-mouth turns negative.

- 3** A firmware OTA update pushed to 500,000 devices in the field reduces field failure rate from 3% to 1.5% by correcting a software-induced hardware stress condition. Model the annual financial impact of that improvement. How do you present that ROI to a leadership team that needs to prioritize the firmware engineering resources required?

Annual savings: $(3\% - 1.5\%) \times 500,000 \text{ units} \times (\text{replacement cost} + \text{shipping} + \text{support}) = \text{significant}$. At \$55 per incident, the improvement saves approximately \$4.1M/year. That ROI case is straightforward to make if the model is built correctly. The PM who can make this calculation has access to engineering resources that the PM who says 'quality matters' does not.

GRADUATE / MASTER'S QUESTIONS

- G1** Juran's quality engineering distinguishes between prevention costs (designing quality in), appraisal costs (testing quality out), and failure costs (internal and external). For a hardware product, where is each dollar of quality investment most productive? How does that change across the product lifecycle from EVT to field maturity?
- G2** Reinertsen argues that queue management is the most important and least-practiced skill in product development. How does that argument apply to the field return queue — the accumulated backlog of units awaiting diagnosis, repair, or replacement? What PM practices keep the return queue from becoming a cash flow crisis?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often treat field failure rate as an engineering parameter that the PM receives, not one the PM

influences. The PM influences field failure rate through: PRD reliability requirements, DVT acceptance criteria, packaging design (which affects shipping damage rate), and OTA strategies that correct software-induced hardware stress.

The assumption that the warranty accrual rate can be set at 1-2% of revenue without modeling it from first principles is one of the most common and costly mistakes in hardware financial planning. Students must build the model from actual failure rate and cost data, not from industry averages.

Graduate engineering students sometimes conflate MTBF with product lifetime. MTBF is a statistical measure of failure rate, not a guarantee. A product with an MTBF of 10,000 hours does not mean each unit will last 10,000 hours — it means that in a large population of units, the average time between failures is 10,000 hours. Some units will fail much sooner.

CONNECTIONS TO OTHER SESSIONS

The hardware economics model is the deliverable that most directly informs the final pitch (Week 15). The model connects to every previous technical session: BOM (Week 3), firmware OTA savings (Week 6), supply chain resilience cost (Week 7), manufacturing yield (Week 10). By Week 15, students should be able to articulate the full margin waterfall from retail price to net margin.

FACILITATOR NOTES

The model workshop requires that students have their BOM inputs from Week 3 and have updated them based on the Week 10 manufacturing discussion. Send a reminder before class. The most common workshop failure mode: students who haven't done the BOM work spend the workshop filling in the BOM instead of modeling the reliability economics. Pre-screen BOM completeness before class and redirect students who need to catch up.

Week 12 · Enterprise Hardware

FORMAT

Case + role-play

TIMING

75 min

LEVEL

MBA + Grad Engineering

SESSION OVERVIEW

B2B hardware product management differs from consumer hardware in nearly every dimension: the buyer is not the user, the sales cycle is 6–18 months, the procurement process involves IT security review, legal negotiation, and capital budgeting, and the support obligation is a service-level agreement rather than a warranty. This session uses a B2B hardware case to expose those differences and equips students with the PM practices that make enterprise hardware products commercially successful.

THEORETICAL FRAME

The Copenhagen Wheel's pivot to fleet management (B2B) is the course's primary B2B case, but it's supplemented here by a broader look at enterprise hardware procurement: what a government or corporate IT department actually requires, how those requirements differ from consumer spec, and what a PM must include in the PRD to make an enterprise sale possible.

SUGGESTED DISCUSSION ARC

0–10 min

Ask: 'Who makes the decision to buy 500 e-bikes for a city fleet? What does that person care about that a consumer buyer doesn't?' Surface: total cost of ownership (TCO), not unit price; service-level agreements (SLAs), not consumer warranty; IT security clearance, not unboxing experience; procurement compliance, not impulse purchase.

10–25 min

Lecture: five ways B2B hardware PM differs from consumer. (1) The buyer is not the user — the fleet manager buys; the rider uses. (2) Procurement cycles are long and bureaucratic — 6–18 months from interest to purchase order. (3) IT security is a gate, not a feature — enterprise connected hardware must pass security review before deployment. (4) SLAs are contractual — uptime commitments, support response times, and replacement timelines are in the contract. (5) Fleet management is a product requirement — the buyer needs to manage, monitor, and maintain hundreds or thousands of units.

25–55 min

Role-play: pairs play a hardware startup PM presenting to a city fleet manager evaluating a 300-unit micromobility pilot. The city manager has a list of procurement requirements (distributed in advance). The PM must demonstrate how their product meets each requirement. Debrief using Questions 1–3.

55–70 min

Graduate section: Foote, Shapiro/Moriarty, Utterback. Questions G1 and G2.

70–75 min

Close: students add an enterprise requirements section to their PRD if their product has B2B applicability. 'Every connected hardware product has a potential B2B buyer. Knowing what that buyer requires is a product design input, not just a sales concern.'

DISCUSSION QUESTIONS — ALL STUDENTS

- 1** A city government wants to deploy 500 connected scooters for a municipal fleet. What requirements will the procurement process impose on the hardware product that would not appear in a consumer product spec? List at least six.

Expected answers: (1) FedRAMP or equivalent security certification for cloud data. (2) FIPS 140-2 encryption for data at rest and in transit. (3) Fleet management dashboard accessible to city IT staff. (4) SLA: 95% fleet availability during operating hours. (5) MTTR (mean time to repair) of less than 24 hours for any hardware failure. (6) Domestic manufacturing requirement. (7) ADA accessibility compliance. (8) Open data standard for reporting.

- 2** The enterprise procurement cycle for a \$500,000 fleet hardware contract typically involves 6 phases: awareness, RFI (request for information), RFP (request for proposal), evaluation, negotiation, and contract. What should the PM be doing to support the sales process at each phase? Which phases are product problems and which are sales problems?

Product problems: awareness (does the product's feature set attract the right buyers?), RFI (do the published specs answer the questions enterprise buyers ask?), RFP (is the product capable of meeting the stated requirements?), evaluation (does a pilot deployment perform to spec?). Sales problems: negotiation and contract. The PM owns the technical case; sales owns the commercial case.

- 3** A fleet operator reports that 8% of units in the first deployment are offline at any given time — firmware connectivity issues on their enterprise WiFi network. Your consumer product has a 1% offline rate on home WiFi. What does this failure reveal about the PM's responsibility when selling hardware into enterprise IT environments?

Enterprise WiFi environments (802.1X, VLAN segmentation, captive portals, firewall rules) are fundamentally different from home WiFi. The PM should have specified enterprise WiFi compatibility testing as a DVT requirement and certified compatibility with the most common enterprise network configurations before launch.

GRADUATE / MASTER'S QUESTIONS

- G1** Utterback describes how competitive dynamics shift as a product reaches dominant design — from product innovation to process and cost innovation. How does that shift manifest in B2B hardware markets? What is the dominant design dynamic for micromobility fleet hardware?

- G2** The Challenger Sale framework describes B2B sales success as coming from teaching customers something new about their business, tailoring the message to the specific buyer, and taking control of the sales conversation. How does the PM enable that kind of sales motion? What does the PM need to provide that a product brochure cannot?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often treat enterprise requirements as edge cases or special features. In many hardware markets, the enterprise buyer (city government, healthcare system, logistics operator) is the primary revenue driver, and consumer sales are secondary. The PM who designs for the consumer and treats enterprise as an afterthought has the priorities backward.

The 'SLA is a sales promise' misconception: SLAs are product requirements. If the product cannot deliver 99% uptime, a sales team that commits to 99% uptime in a contract creates a product liability. The PM must validate that the product architecture can support the SLA before sales commits to it.

Graduate engineering students may treat enterprise security requirements as bureaucratic overhead. These requirements exist because enterprise IT departments are accountable for breaches that affect thousands of users. The security requirements are not arbitrary — they reflect real threat models.

CONNECTIONS TO OTHER SESSIONS

Enterprise hardware connects to Week 9 (IoT — enterprise connectivity requirements differ significantly from consumer), Week 10 (manufacturing — enterprise customers often have supply chain transparency requirements), and Week 13 (platform strategy — enterprise hardware platforms create the deepest lock-in). The fleet management dashboard required by enterprise buyers is itself a platform product.

FACILITATOR NOTES

The role-play works best when the 'city fleet manager' role is played by a student with urban planning or government background, if available. The purchasing requirements list should be drawn from an actual municipal RFP — many are publicly available. San Francisco, Los Angeles, and Denver all have published micromobility RFPs that work well as role-play inputs.

Week 13 · Platform Strategy**FORMAT**

Seminar + case analysis

TIMING

75 min

LEVEL

MBA + Grad Engineering

SESSION OVERVIEW

The most durable hardware businesses are platform businesses: products that become the foundation on which other products, services, and ecosystems are built. Apple's iPhone is the canonical example, but the pattern appears in hardware at every scale: Peloton built a fitness platform on exercise hardware; JUMP built a mobility platform on connected bikes; Nest became the hub of a home automation ecosystem before being absorbed into Google Home. This session teaches students to distinguish between a hardware product and a hardware platform, and to ask whether their product concept has platform potential.

THEORETICAL FRAME

Platform economics in hardware differ from pure digital platforms in one critical way: the physical product creates switching costs that digital platforms cannot. A customer who has bought a Peloton and has three years of workout data, six accessories, and a social network attached to the device has switching costs that Spotify simply cannot create. That stickiness is both a business advantage and an ethical question — the session should address both.

SUGGESTED DISCUSSION ARC**0–10 min**

Open with Parker/Van Alstyne's platform definition: a platform facilitates interactions between two or more groups and creates value from those interactions. 'Is the iPhone a platform? Is a Peloton? Is a connected bike?' Map each against the definition. Surface: all three qualify, but the mechanisms of value creation and the groups being connected differ.

10–25 min

Lecture: four hardware platform models. (1) Ecosystem enabler (iPhone App Store): the hardware enables third-party software developers to create value. (2) Data platform (Nest/Google Home): the hardware generates data that becomes the platform's primary value. (3) Subscription hardware (Peloton): the hardware is the customer acquisition channel for a recurring revenue service. (4) Fleet-as-platform (JUMP/Superpedestrian LINK): the hardware fleet is the platform on which a mobility service is built.

25–55 min

Case analysis: students analyze their own product concept against the four platform models. Can their product generate network effects? What would a third-party ecosystem look like? What recurring revenue could the hardware support? Questions 1–3.

55–70 min

Graduate section: Cusumano/Gawer, Rochet/Tirole, Zhu/lansiti. Questions G1 and G2. The Zhu/lansiti failure factors are the most useful for students who want to understand platform risk.

70–75 min

Close: 'Your hardware product is either a platform, a product that could become a platform, or a product that intentionally isn't a platform. Know which one yours is and why.'

DISCUSSION QUESTIONS — ALL STUDENTS

- 1 **Peloton's business model is described as 'hardware as a subscription delivery mechanism' — the bike loses money at the hardware margin but earns subscription revenue over a 3–5 year membership lifecycle. Model the Peloton unit economics: at what subscription duration does the customer become profitable? What does that imply for the hardware design decision — how long must the hardware last?**
Peloton historically: bike ~\$2,000 retail, hardware margin negative at launch. Subscription \$44/month. At \$44/month, breakeven on hardware in roughly 18–24 months. Hardware must last at least 3 years for the business model to work. This is a hardware reliability requirement derived from the business model — not an engineering choice.
- 2 **Apple's platform strategy for iPhone creates switching costs through: accumulated app purchases, device ecosystem (AirPods, Apple Watch, iCloud), and social features tied to Apple ID. What switching costs does a connected bike or scooter platform create? Which of those switching costs are created by the hardware, and which by the software?**
Hardware-created switching costs: physical installation, accessories, ergonomic settings. Software-created: workout history, route data, social features, subscription. The deeper the software switching costs, the more durable the platform — but hardware platform businesses often underinvest in software because the hardware is the visible product.
- 3 **Hoffman's @arcktip essay argues that pricing is product design. In the context of a hardware platform, how does pricing strategy shape who the platform attracts, what behaviors it incentivizes, and what ecosystems can be built on top of it?**
Pricing shapes the two-sided market: low hardware prices attract more users (good for network effects) but reduce margin (bad for sustainability). Subscription pricing creates predictable revenue but may deter price-sensitive users. Open APIs attract developer ecosystems but reduce platform control. Each pricing decision is a platform architecture decision.

GRADUATE / MASTER'S QUESTIONS

- G1 **Gawer and Cusumano describe 'platform leadership' as a distinct strategic capability: the ability to manage the tension between controlling the core platform and enabling third-party ecosystem value creation. What decisions does a hardware PM make that constitute platform leadership? What does a hardware platform PM own that a hardware product PM does not?**
- G2 **Rochet and Tirole's two-sided market model describes the pricing structure of platforms: the platform charges different prices to each side to maximize total value creation. For a micromobility fleet platform (hardware deployed by operators, used by riders, regulated by cities), how many 'sides' are there, and how should the platform price each? What happens when the pricing structure that works for riders conflicts with what operators or cities will accept?**

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often assume that being a 'platform' is always the right strategic choice. Platforms require network effects to generate value; if the product category doesn't have strong network effects, platform strategy is a liability (complexity without benefit). Not every hardware product should try to be a platform.

The razor-and-blades model is often misapplied: it requires that the blades (consumables or subscriptions) be genuinely proprietary and tied to the razor (hardware). Peloton's model works because the subscription content is exclusive. A hardware product with commodity subscription content cannot sustain the model.

Graduate engineering students sometimes think of platform APIs as a purely technical decision. API strategy is a platform governance decision: what third parties can build, at what cost, with what access to data, and under what terms. The PM owns the API strategy, not the engineering team.

CONNECTIONS TO OTHER SESSIONS

Platform strategy is the culminating analytical session before the final pitch. The four platform models introduced here give students a vocabulary for articulating their product's business model in the Week 15 pitch. Platform strategy also connects back to Week 3 (BOM economics — platform models often involve subsidizing hardware) and Week 9 (IoT — connectivity is the mechanism of platform value creation).

FACILITATOR NOTES

The Peloton unit economics exercise works well as a 5-minute individual exercise before group discussion. Students who have done the hardware economics model (Week 11) can apply that framework here; students who haven't feel the gap. Use it as a diagnostic and as motivation for the homework.

Week 14 · Sunset & Obsolescence

FORMAT

Seminar + policy workshop

TIMING

75 min

LEVEL

MBA + Grad Engineering

SESSION OVERVIEW

Hardware product management doesn't end at launch — it ends when the last unit is decommissioned. End-of-life planning is among the least-taught and most consequential skills in hardware PM: what happens when a cloud service that your 2019 hardware depends on is discontinued? What are your legal obligations when e-waste from your product turns up in a Ghanaian landfill? What does right-to-repair mean for your product, and how should you respond to it? This session addresses all three questions and requires students to add an EOL plan to their PRD.

THEORETICAL FRAME

Crawford's 'Atlas of AI' provides the ethical grounding: the material reality of hardware products — rare earth mining, Shenzhen assembly lines, e-waste processing — is invisible to most PM and MBA students. Making it visible changes the ethical calculus of hardware product decisions. Perzanowski and Schultz's 'End of Ownership' provides the legal framework: what does a customer actually own when they buy a hardware product, and what happens when the manufacturer discontinues support?

SUGGESTED DISCUSSION ARC

0–10 min

Open with a concrete scenario: 'You are the PM of a connected thermostat company. Your company has just been acquired. The acquirer decides to shut down the cloud backend that your 2019 thermostats depend on. 200,000 thermostats become less functional. What are your legal obligations? Your ethical obligations? Your product obligations?' Surface: this is not a hypothetical — it has happened multiple times (Revolv, Wink, Nest guard).

10–25 min

Lecture: four EOL challenges. (1) Software/cloud dependency: hardware that depends on cloud services becomes less functional when those services end. PM must plan for cloud service lifecycle in the PRD. (2) Right-to-repair: EU and US regulations increasingly require manufacturers to provide spare parts, tools, and documentation for repair. PM must design for repairability. (3) E-waste obligations: RoHS and WEEE (EU), state-level e-waste laws (US), extended producer responsibility. (4) Firmware end-of-life: security patches must continue as long as units are in the field; ending security patches creates liability.

25–55 min

Policy workshop: pairs draft a 'product end-of-life plan' for their product concept. The plan must address: (1) minimum cloud service commitment, (2) data portability for customers at EOL, (3) spare parts availability commitment, (4) firmware security patch timeline, (5) recycling program. Questions 1–3.

55–70 min

Graduate section: Stahel, MacKenzie, Mazzucato. Questions G1 and G2.

70–75 min

Close: students add the EOL section to their PRD. 'A PRD that doesn't address EOL has not finished thinking about the product.'

DISCUSSION QUESTIONS — ALL STUDENTS

- 1 **In 2016, Google's Nest acquired Revolv, a smart home hub company, and 16 months later shut down the Revolv cloud service, permanently bricking every Revolv device. What obligations should a hardware company have to customers who purchased a cloud-dependent product? Draft three principles for an ethical hardware EOL policy.**
Expected principles: (1) Minimum operational commitment: the company commits to maintaining core functionality for X years after purchase. (2) Advance notice: customers receive Y months' notice before any service change that affects product functionality. (3) Local mode: the product is designed with a local fallback mode that maintains basic functionality if cloud services end.
- 2 **The EU Right to Repair Directive (2023) requires manufacturers of certain product categories to provide spare parts, tools, and repair documentation for a minimum period after sale. If your product is subject to this directive, how does it change your product design decisions? Your BOM? Your supply chain?**
Design changes: modular design that allows component-level replacement rather than whole-unit replacement. BOM changes: components must be sourced from suppliers who will maintain spare part supply for 10 years. Supply chain changes: spare parts inventory must be maintained long after production ends. These are significant engineering and supply chain constraints that affect the product's cost structure from day one.
- 3 **Crawford's 'Atlas of AI' traces the material supply chain of AI and connected products from rare earth mining to e-waste processing. What does that supply chain reveal about the true cost of a hardware product? Which of those costs appear in your BOM, and which are externalized?**
Most environmental costs are externalized: rare earth mining impacts, Shenzhen assembly worker conditions, and e-waste processing in informal recycling markets in Ghana or India are not in the BOM. Extended producer responsibility regulations are beginning to internalize some of these costs. The PM who understands the full supply chain can anticipate regulatory changes before they arrive.

GRADUATE / MASTER'S QUESTIONS

- G1 **Stahel's circular economy framework describes a product lifecycle in which materials are kept in use at their highest value for as long as possible. Apply that framework to a connected hardware product. What product design decisions would maximize circularity? How does circular design conflict with the design choices that maximize initial performance and minimize BOM?**
- G2 **Perzanowski and Schultz argue that the legal concept of 'product ownership' has been eroded by software license agreements and cloud dependency — customers increasingly rent rather than own the products they buy. Is that erosion good or bad for hardware product companies? For customers? For society? What regulatory response is appropriate?**

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

EOL planning is not a legal compliance exercise — it is a product design exercise. Products that are designed for EOL (modular, repairable, with clear data portability) are more trusted by customers, more compliant with emerging regulation, and less expensive to manage at the end of their life.

Students often assume that EOL is the last thing to think about. In hardware, EOL decisions must be made at the PRD stage because they affect component selection (spare parts availability), design (modularity for repair), supply chain (long-term supplier commitments), and software architecture (local mode capability).

Graduate engineering students sometimes treat e-waste as a regulatory problem rather than a design problem. The products that generate the least e-waste are the ones designed for longevity, repairability, and material recovery from the first sketch — not the ones that get e-waste regulations added to their spec sheet after the design is locked.

CONNECTIONS TO OTHER SESSIONS

EOL is the closing chapter of the course's narrative arc: from 'why hardware is different' (Week 1) to 'what done means for a physical product' (Week 14). The EOL plan added to the PRD in this session completes the PRD that will be presented in Week 15. The Crawford reading connects back to Week 9 (IoT — the data centers and cloud infrastructure that IoT depends on have significant material footprints).

FACILITATOR NOTES

The Revolv case is particularly effective because students often have personal experience with cloud-dependent devices and have never considered what happens when the cloud goes away. Ask: 'Has any device you own ever stopped working because a cloud service was discontinued or a company was acquired?' Surprisingly many students have this experience. It makes the EOL question immediate and personal.

Week 15 · Final Pitch

FORMAT	TIMING	LEVEL
Public pitch + external panel	120 min	MBA + Grad Engineering

SESSION OVERVIEW

The final pitch is the course's synthesis event. Teams present a complete physical-digital product concept: a PRD, a hardware roadmap with explicit decision gates, a BOM estimate with margin model, a dependency map, a supply chain strategy, a firmware and IoT architecture summary, and a go-to-market plan — all to a panel of hardware practitioners. The evaluation criteria reflect the course's full technical and business framework: does the team understand what building this product actually requires?

THEORETICAL FRAME

No new theory. The session tests whether students can synthesize the course's toolkit — hardware economics, systems thinking, build cycle management, firmware strategy, supply chain planning, IoT architecture, platform strategy, and EOL planning — into a coherent and credible product proposition. The best pitches are honest about what they don't yet know.

SUGGESTED DISCUSSION ARC

0–5 min	Frame: 'This is not a startup pitch — it is a product proposal to a hardware PM review board. The question is not whether investors would fund this. The question is whether this team understands what building this product actually requires — the hard parts, the constraints, and the decisions they haven't made yet.'
5–95 min	Presentations: each team presents for 15 minutes + 10 minutes panel Q&A. Panel should be composed of practitioners: a hardware PM from consumer electronics, a contract manufacturing specialist, and a hardware investor or product operations leader.
95–115 min	Panel synthesis: what was the strongest PRD section across all presentations? What was the most common gap? Which team most clearly understood the relationship between BOM and business model?
115–120 min	Instructor close: return to Week 1's seven ways hardware PM differs from software PM. 'Which of these seven dimensions did your product concept require you to think hardest about? Which are you still not sure you understand?'

DISCUSSION QUESTIONS — ALL STUDENTS

- 1 What is the single most expensive decision in this product's development — the decision that, if wrong, would be hardest to reverse? How have you managed the risk of

being wrong about it?

For most hardware products: the tooling commitment (NRE investment in manufacturing molds). The right answer involves a decision gate: the team doesn't commit to tooling until they have validated the product's core value proposition with a DVT-quality prototype.

2 **Your BOM estimate assumes 100,000 units per year at year two. Walk me through the margin waterfall: from retail price to gross margin after warranty. At what volume does this business become viable?**

Students must walk through: retail price → channel margin → ASP to company → COGS (BOM + manufacturing) → gross margin before warranty → warranty/support expense → net gross margin. A team that cannot do this calculation has not finished the business model.

3 **Which build stage is your product at right now — POC, EVT, DVT, or PVT? What must you prove at the next build stage before committing to the one after it?**

Most student concepts are at POC or early EVT conceptually. The right answer involves explicit criteria: 'We are at the POC stage. Before EVT, we need to validate [specific technical risk] with a functional prototype and confirm [specific market assumption] with [specific evidence].'

GRADUATE / MASTER'S QUESTIONS

G1 If this product succeeds and reaches 1 million units in the field, what does its end-of-life look like? How long will you support it, and what happens to the hardware when support ends?

G2 Looking back at the course: which concept — BOM economics, dependency mapping, EVT/DVT/PVT, OTA strategy, supply chain resilience, platform strategy, or EOL planning — was most different from what you expected before the course? What changed in how you think about physical products?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Teams sometimes present polished slides rather than a rigorous product proposal. The panel should be instructed to ask for the BOM breakdown, the roadmap decision gates, and the firmware rollout strategy — the sessions where the hardest thinking is required. Slide quality is not a substitute for analytical rigor.

The most common final pitch failure mode: teams present a consumer electronics concept without having thought through the manufacturing side. A team that can describe the user experience in detail but cannot state the target BOM, the CM strategy, or the build cycle sequence has not done the work this course requires.

CONNECTIONS TO OTHER SESSIONS

The final pitch is the destination of the entire course. Every deliverable (dependency map, BOM workshop, roadmap, PRD, economics model) feeds into it. Students who have maintained their deliverables throughout the course find the final pitch a synthesis exercise; students who haven't find it a crisis. The instructor's job in the final weeks is to ensure the former and prevent the latter.

FACILITATOR NOTES

Panel composition is critical. The most valuable panelists: (1) A hardware PM who has personally managed EVT/DVT/PVT cycles — they can ask the most precise build-stage questions. (2) A CM specialist or manufacturing engineer — they can challenge the BOM and yield rate assumptions immediately. (3) A hardware investor — they can challenge the business model math. Brief all panel members with the course's seven-point framework from Week 1. The best final sessions feel like a rigorous product review, not a pitch competition.

APPENDIX: QUICK-REFERENCE QUESTION BANK BY THEME

These questions can redirect or deepen discussion in any session:

Urban Systems (any session)

- What does this product do to the urban system? What feedback loops does it create?
- What would Jacobs say about this product? What would Bertaud say?
- What does optimizing for the measurable sacrifice that is not measurable?

Equity & Justice (Weeks 5, 8, 11, 13)

- Who benefits from this product in year one? In year five? Are they the same people?
- Which equity failure mode applies here: design, deployment, or displacement?
- What would Sheller's mobility justice framework require of this product as a condition of legitimacy?

Regulation & Governance (Weeks 2, 9, 14)

- Is this regulation alleviating an externality or redesigning the city? What is the difference in this case?
- What does the regulatory text reveal about what the city actually values?
- What would a collaborative — rather than adversarial — regulatory strategy look like?

Data & Measurement (Weeks 3, 10)

- What does this product measure, and what does optimizing for that metric make worse?
- Who generated this data, who owns it, and who can see it?
- What would Loukissas say about applying this dataset in a different city context?

Business Model & Capital (Weeks 6, 12, 14)

- Who internalizes the profits and who absorbs the externalized costs?
- What happens to this product's mission if it is acquired by a larger platform?
- What is the minimum business model that could sustain this product without extracting value from the city?

Practitioner Perspective (Weeks 5, 13, 15)

- What does Patrick's account of building JUMP see that academic readings cannot? What does it miss?
- What would you do differently from the SoBi/JUMP arc, given what you now know?
- If you were advising a city council on whether to permit this product, what three questions would you ask the product team?

This guide is a living document. Faculty who use it are encouraged to annotate, revise, and return observations to Patrick Hoffman (@arcktip) so that future iterations benefit from classroom experience across institutions.